Fast and Accurate post-layout simulation methodology using 4nm LVS rule deck

Woonggyu Lee
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
wg21.lee@samsung.com

Jihoon Park
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
jh8504.park@samsung.com

Jaehong Park
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
j0127.park@samsung.com

Yongjae Park
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
youngjae.park@samsung.com

Minkyung Kim
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
mk723.kim@samsung.com

Taehyung Lee
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
tjhthnet.lee@samsung.com

Hyojin Kim
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
hyojin03.kim@samsung.com

Changyoon Shin
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
cy0921.shin@samsung.com

Jiseon Lee
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
jiseons.lee@samsung.com

Seungjae Jung
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
seungjae77.jung@samsung.com

Bonhyuck Koo
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
bonhyuck.koo@samsung.com

Sungjae Lee
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
sung-jae.lee@samsung.com

Jongwook Kye
Samsung Foundry
Samsung Electronics
Gyeonggi-do, South Korea
jongwook.kye@samsung.com

2022 IEEE 31st Microelectronics Design & Test Symposium (MDTS) | 978-1-6654-6816-9/22/$31.00 ©2022 IEEE | DOI: 10.1109/MDTS54894.2022.9826968

***Abstract*— As the feature size decreases in the IC design flow, fast and accurate post-layout simulation is becoming increasingly important to meet tight design completion cycle requirement. When the designer performs the simulation of the sub block by reflecting the parasitic of the top, it may take several days. Therefore, various methods such as skip-cell and RC reduction are implemented to reduce simulation time for large IP or a small chip. While those methods are used to reduce runtime, accuracy may suffer compared to general simulation results. This paper addresses new method of selectively extracting devices required for simulation and maintaining the layout parasitic RC using the LVS (layout vs schematic) rule file to satisfy both fast runtime and accuracy. As a result of applying this method to 4nm ESD simulation, the post-simulation runtime has been dramatically decreased from about 16 hours to 40 minutes, and when checking the simulation result, accuracy is also guaranteed by less than 1.5% from the reference value.**

Keywords—LVS, PDK, Post-Layout simulation

I. INTRODUCTION

The post-layout simulation for analog and mixed signal circuit becomes more and more challenging for the run-time due to increasing number of devices, nets and parasitic Resistance and Capacitance (RC) components from Parasitic Extraction (PEX) in advanced technologies [1]. When running post-layout simulation with a large block, simulation including all cells may take several days or it may not be even possible to finish the run and check the results.

There are three methods used so far to reduce post-layout simulation runtime.

1. *Method for reducing post-layout circuits by sparsity-aware SIP (Sparse Implicit Projection) reduction [2]*
2. *Simulation and analysis using fast-spice simulators [3]*
3. *Ignore the parasitic RC of unnecessary user-defined cells (skip-cell method)*

However, all of the above methods require too much time to perform post-layout simulations, and cannot guarantee accurate results due to forced parasitic RC reduction. All of parasitic extraction and simulation tools support the option of reducing parasitic RC in the complex network and users have difficulty adjusting and using this option appropriately for acquiring the simulation result with appropriate accuracy. The transistor level simulation with a lot of devices and nets including parasitic components from a flat netlist is always very complex and time consuming, sometimes impossible [4].

Due to the problem that it takes too long to proceed with ESD post-layout simulation, designers have a problem of sign-off with inaccurate results from pre-simulation or conventional methods that have been used to reduce runtime.

This paper presents a method that can maintain accuracy while dramatically reducing simulation runtime using PDK

978-1-6654-6816-9/22/$31.00 ©2022 IEEE

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:37:26 UTC from IEEE Xplore. Restrictions apply.

(Process Design Kit) LVS (Layout vs Schematic) rule deck without having to manipulate circuit and layout design. The new method has initially been verified with 4nm ESD post-layout simulation, and has achieved excellent results in both simulation runtime and accuracy.

II. EXPERIMENTAL CHALLENGES

Using the traditional method, it has been confirmed that simulation runtime and accuracy have a trade-off relationship with each other. Therefore, we study how to solve both runtime and accuracy by identifying what is the biggest factor in runtime and what factors affect accuracy.

A. Runtime Reduction

When a circuit designer only wants to perform and review the functional behavior of a specific circuit quickly, all of transistors shouldn't be required for the post-layout simulation. Therefore, we have explored the method to only extract activated devices on the entire circuit required to the post-layout simulation in order to confirm functional behaviors. As a result of checking various factors that affect runtime, the most important factor is the number of devices, especially the number of FET devices. We checked how the number of devices and runtime change while removing cells that do not affect ESD simulation from the reference DB. Figure 1 shows the correlation with simulation runtime depending on the number of devices. It can be confirmed that the simulation runtime decreases as the number of devices decreases.

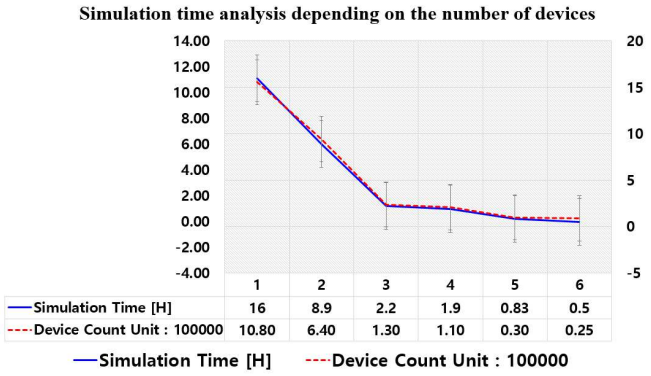


Fig. 1. Simulation time analysis depending on the number of devices. The red dotted line shows the number of devices, and the blue line shows the simulation time.

One of the introduced methods about reducing the number of devices is limited to memory designs which are relatively a simple array structure [5] and it has the disadvantage in that designer needs to modify the layout. Our method extracts only the selective devices without the layout modification through the LVS step by the circuit or layout designer. Then, we can get the results of the simulation with the selective devices. This method is the fastest way to check the functional impact on top-level designs that take a long time or cannot perform the post-layout simulation.

B. Simulation Accuracy

The post-layout simulation runtime can be definitely reduced by using the conventional skip-cell method. However, since all parasitic RCs disappear, it is not possible to check the parasitic effects that affect major critical devices or power signals. In other words, accuracy is not guaranteed. Therefore, if we extract all parasitic RC parameters generated by the layout effect while reducing the number of devices, we expected the result of the smallest difference compared to the reference result. That is, it can be assumed that good results can be obtained in terms of accuracy if all activated devices selected for function check and routing-related parasitic RC are extracted.

Figure 2 shows that the simulation results are different depending on whether or not the parasitic RC is extracted. Compared to simulation results with parasitic RC parameters, it shows no valid Vgs waveform and the reduced Vpad voltage due to absence of parasitic resistance on the PAD. The reliability of the victim device must meet a specific condition ($Vgs \leq |7V|$). If parasitic RC parameters are not included, such as the skip-cell method, $|Vgs| \cong 5.497V$ is confirmed to have no problem because it satisfies the conditions, and the designer determines that there is no problem. However, if the parasitic RC parameter is included, $|Vgs| \cong 7.394V$ is not satisfied with the reliability condition. Therefore, the designer should recognize that there is a reliability problem and review the design.

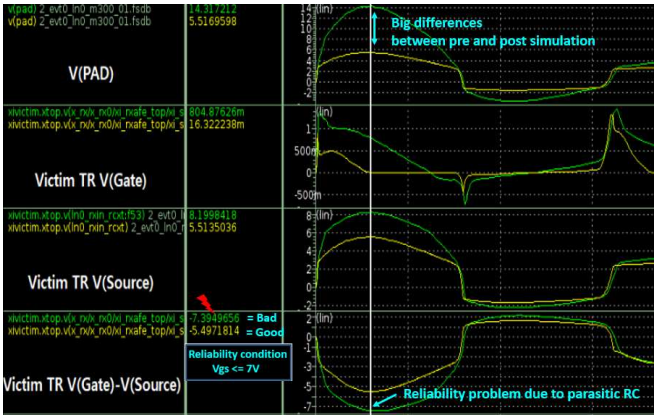


Fig. 2. ESD post-layout simulation. Green shows results with the parasitic RC, and Yellow shows results without the parasitic RC.

The most important point is whether both runtime and accuracy can be solved at the same time. In order to find a way, the experiment was conducted with the aim of solving it using LVS rule deck.

In this paper, our initial experiment and evaluation focused on the Electrostatic Discharge (ESD) circuit simulation with extraction of selected devices related to ESD protection circuits. The ESD protection circuit occupies a small portion of the entire design area and does not require the logic area to review the ESD function operation through results of the post-layout simulation.

III. SELECTIVE EXTRACTION FLOW

SE (Selective Extraction) is a new methodology, which selectively extracts only the devices of the cells designer wants to evaluate. The most significant factor to shorten runtime is to reduce the number of devices that needs to be simulated as shown in Figure 1. The LVS rule deck can extract devices by defining only the cells desired by the designer. In addition, since accuracy should not be affected for the runtime reduction, the parasitic RC should be maintained not only in selected cells but also in unselected cells. Therefore, the simulation is finally performed by full parasitic RC and selective devices extracted by LVS rule deck with SE method

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:37:26 UTC from IEEE Xplore. Restrictions apply.

as shown in Figure 3. This new methodology is very effective when simulation is performed using some areas or devices rather than situations in which most areas or devices are required from top design.

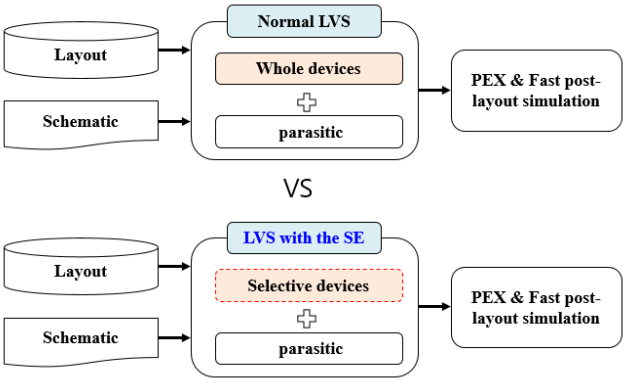


Fig. 3. Selective Extraction Flow

A. Selection of schematic cells on the circuit

The selected cell should include all devices associated with the review point through the post-layout simulation. The missing device can produce a post-layout simulation result with poor accuracy, so the cell list must be redefined correctly.

B. Layout cells corresponding to the selected schematic cells

Designers should know exactly where their circuit designs are located in the layout and also know exactly where switching devices are related to victim devices. The selective cells should be examined from the layout to ensure that the area does not affect the desired simulation.

C. Factors that determine the 'smart' selection of cells that drive a smaller number of devices

It is recommended to select the 'smart' selection determination method of cells that drive a smaller number of devices depending on the situation.

1. The first method is to select cells that contain the necessary devices from the layout. It is confirmed that there is no big difficulty in checking the cell area of most designers themselves in the layout.

2. The second method is to select the schematic based selective cell. Regardless of the layout situation, the designer selects the cells required for simulation in the schematic design or the cells performed in the previously used pre-simulation.

3. Finally, only the devices in the critical path are selected. In this way, the fewest devices will be selected.

Figure 4 highlights the selected areas through the above-mentioned three methods. If the selected area is small, faster simulation can be performed, but removing devices close to the critical path can affect accuracy. The designer can proceed with the 'smart' selection mentioned above according to various situations.

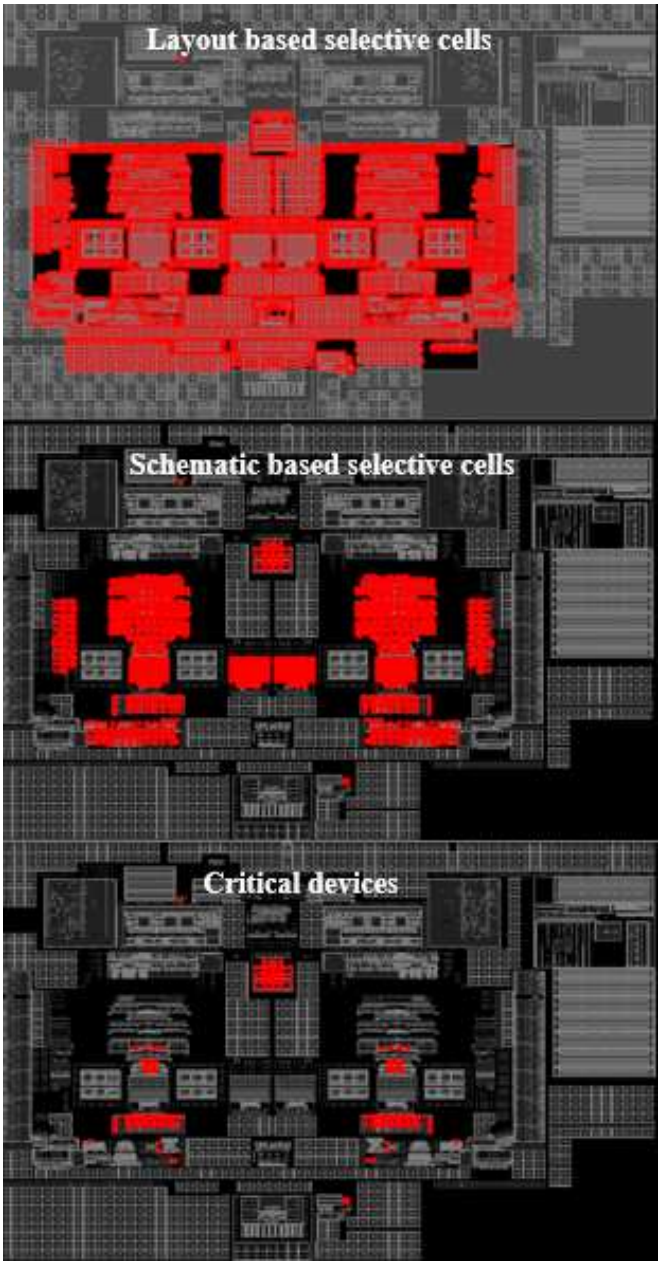


Fig. 4. Layout based selective cells vs Schematic based selective cell vs Critical devices selection.

D. LVS rule deck modification for SE

All devices should be extracted for cells selected by the d esigner, and NON-FET devices should be extracted for unse lected cells for accurate simulation. Since the power/ground parasitic RC or passive devices are important in whole cells [6], only the FET devices are not extracted by the LVS in the unselected area as shown in Figure 5.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:37:26 UTC from IEEE Xplore. Restrictions apply.

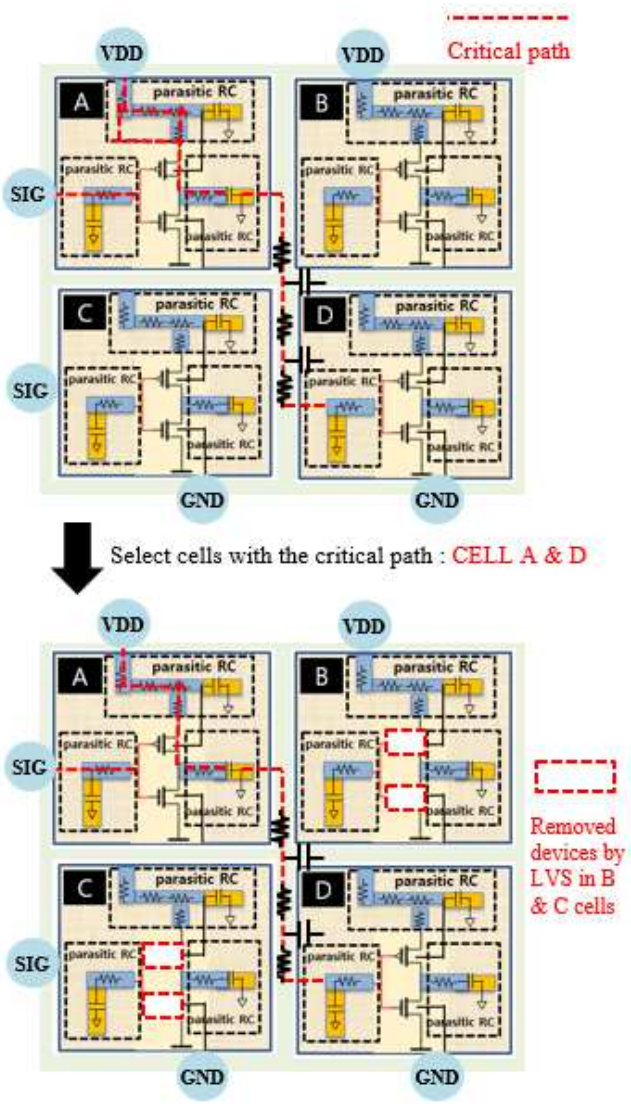


Fig. 5. LVS rule operation with SE methodology

E. PEX and Simulation

Normal PEX and simulation can be proceed using devices and parasitic derived layers retrieved from the LVS for SE. Designers can obtain fast and accurate results without requiring any layout modifications or other additional work to perform simulation.

IV. EXPERIMENTAL RESULTS

A. Portion of design area for selective cells (ratio)

The measurement of Table1 below is performed for determining how many areas the selective cells occupy in the design approximately. The ratio of selective cells to entire area is not directly related to reduction of extracted devices and could not have proportional expectations for the performance improvement. However, it can be used as an indicator to check whether the designer has defined the selective cells well or not.

TABLE I. AREA RATIO (SELECTIVE-CELLS/ENTIRE REGION)

	Flow	*Selective cell count*	*Area Region [um^2]*	*Area Ratio*
Case1	Conventional	-	326,000	1
	Selective	5	126,699	0.39
Case2	Conventional	-	123,660	1
	Selective	17	84,575	0.68

B. Number of extracted devices through the LVS

To reduce runtime, PEX options to reduce the Parasitic RC nodes are not enough. The most important factor to reduce the runtime of post-layout simulation is the count of devices. As the number of devices used in the simulation decreases, the runtime decreases. As shown in Table 2 below, the number of devices dramatically decreases using the new approach.

TABLE II. CONVENTIONAL VS SELECTIVE EXTRACTION RESULTS

	Flow	*LVS Runtime[H]*	*Device Count*	*SPF Size [G]*
Case1	Conventional	0.44	1,088,177	10
	Selective	0.3	110,936	2.1
Case2	Conventional	0.12	132,071	1.2
	Selective	0.03	25,387	0.5

C. Post-layout simulation results

There are many conventional methods to reduce runtime, such as skip-cell. As shown in Table 3 below, it shows that there is a problem with accuracy in the method used to reduce runtime. The SE method includes not only devices that affect the critical path, but also passive devices such as decoupling cap devices so that there is almost no accuracy impact. Simulation including the parasitic RC in the PEX tool, which is used by general designers, is performed as a reference. The SE method does not differ in terms of runtime compared to the skip-cell method, which is one of the fastest methods for runtime, and in terms of accuracy, it can be seen that there is no significant difference compared to reference.

TABLE III. 4NM ESD POST-LAYOUT SIMULATION RESULTS

	PEX option (R-2020.09-SP5)		*Simulation Runtime [H]*	*Vgs [V/%]*	
Case1	Full SPF	RC	16.18	-7.705	Ref
	Skip-Cell	RC	0.48	-7.093	-7.9%
	Selective Extraction	RC	0.67	-7.819	1.4%
Case2	Full SPF	RC	2.14	4.108	Ref
	Skip-Cell	RC	0.22	3.410	-17%
	Selective Extraction	RC	0.27	4.072	0.4%

As shown in Figure 6, it shows that the general simulation and the SE method results are almost identical. Green is the result of normal post-layout simulation and is used as a reference. Yellow is the convention method (Skip-Cell) to reduce runtime, and compared to Skip-Cell results, Blue SE results show that runtime is almost equal and accuracy is much better.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:37:26 UTC from IEEE Xplore. Restrictions apply.

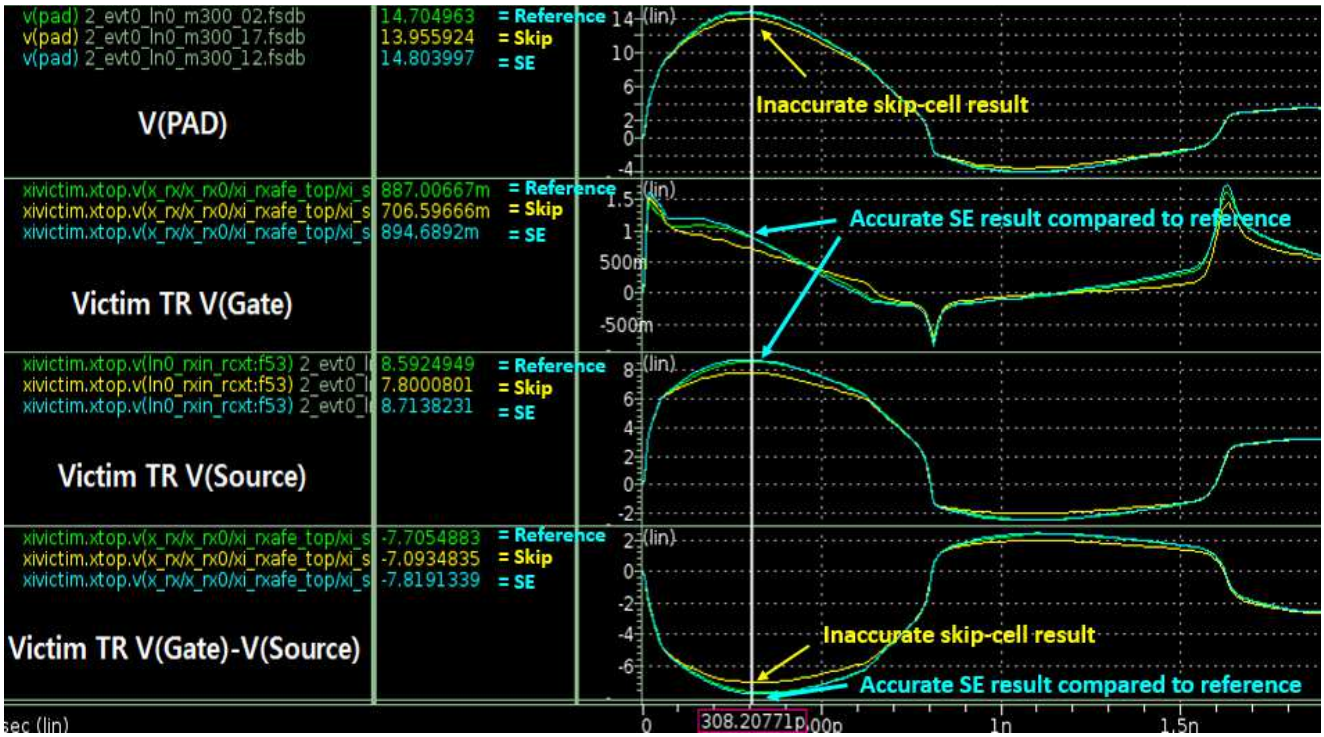


Fig. 6. 4nm FinFET ESD Post-layout simulation: Green indicates the normal post-layout simulation results as reference, Yellow indicates Skip-Cell results, and Blue indicates SE results.

V. CONCLUSION

The methodology discussed in this paper shows that LVS can contribute to fast and accurate post-layout simulation with selective device extraction coding considering designer's fast simulation TAT. The proposed techniques prevent timing lose or economic loss in the faster simulation without breaking the accuracy.

REFERENCES

[1] P. Jay and A. D. Darji, "Analysis of the source/drain parasitic resistance and capacitance depending on geometry of FinFET," 2015 11th Conference on Ph.D. Research in Microelectronics and Electronics (PRIME), 2015, pp. 298-301, doi: 10.1109/PRIME.2015.7251394.

[2] Z. Ye, "Sparsity-aware model reduction for post-layout circuit simulation," 2014 IEEE International Conference on Electron Devices and Solid-State Circuits, 2014, pp. 1-2, doi: 10.1109/EDSSC.2014.7061232

[3] S. Sangameswaran and S. Yamauchi, "Post-layout parasitic verification methodology for mixed-signal designs using fast-SPICE simulators," 2005 IEEE Dallas/CAS Workshop on Architecture, Circuits and Implementtation of SOCs, 2005, pp. 211-214, doi: 10.1109/DCAS.2005.1611173.

[4] P. Daglio, D. Iezzi, D. Rimondi, C. Roma and S. Santapa, "Building the hierarchy from a flat netlist for a fast and accurate post-layout simulation with parasitic components," Proceedings Design, Automation and Test in Europe Conference and Exhibition, 2004, pp. 336-337 Vol.3, doi: 10.1109/DATE.2004.1269268.

[5] Xiaocheng Jing and Ruohe Yao, "A fast-simulation model for post-layout SRAM," 2007 7th International Conference on ASIC, 2007, pp. 1197-1200, doi: 10.1109/ICASIC.2007.4415849.

[6] Ming-Dou Ker and K. . -H. Lin, "ESD protection design for I/O cells with embedded SCR structure as power-rail ESD clamp device in nanoscale CMOS technology," in IEEE Journal of Solid-State Circuits, vol. 40, no. 11, pp. 2329-2338, Nov. 2005, doi: 10.1109/JSSC.2005.857349.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on October 06,2026 at 02:37:26 UTC from IEEE Xplore. Restrictions apply.